ANSWER KEY & MARKING SCHEME SET 1

Chapter 8: **Water Resources (Geography)** | Class 10 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** A is true but R is false.	A true; R false
2	**(b)** reduced storage capacity	Sediment fills the reservoir
3	**(a)** Both A and R are true and R is the correct explanation of A.	Both true; R explains A
4	**(d)** Tehri dam	In Uttarakhand
5	**(c)** rainwater harvesting	Harvesting adds supply
6	**(d)** A is false but R is true.	A false; R true
7	**(a)** local rainwater harvesting	Locally suited
8	**(b)** water is a limited resource	The usable share is small
9	**(a)** Both A and R are true and R is the correct explanation of A.	Both true; R explains A
10	**(a)** Both A and R are true and R is the correct explanation of A.	Both true; R explains A

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	They change river flow and trap silt (1); They block fish migration and flood forests (1).	1+1
12	Water is scarce (1); Efficient methods reduce wastage and make water available for more users (1).	1+1
13	Crops need regular water (1); Scarcity lowers yields and income, and forces farmers to deepen wells (1).	1+1
14	Silt reduces storage so floods spill over (1); Sudden release of water from full dams floods downstream areas (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	People need water as a right (1); Disputes arise over who controls it (1); Fair access should be ensured for all (1).	1+1+1
16	The law made rooftop harvesting compulsory (1); Strict enforcement and awareness helped (1); Groundwater levels rose in Chennai and other places (1).	1+1+1
17	Tankas store water in dry areas (1); Kuls use mountain streams (1); Inundation channels use floods (1).	1+1+1
	OR: Benefits of dams reach some regions only (1); Displaced people may not gain (1); Inequalities between regions can grow (1).	1+1+1
18	It uses local bamboo and spring water (1); Water is delivered drop by drop (1); It needs no energy and wastes little (1).	1+1+1
	OR: High-yielding seeds need more irrigation (1); Farmers use tube wells (1); Water tables fell (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Multipurpose projects store water behind dams for irrigation, hydroelectric power, water supply and flood control, as at Bhakra-Nangal on the Sutlej and Hirakud on the Mahanadi (1); Jawaharlal Nehru called dams the temples of modern India because they promised agricultural and industrial growth (1); The costs include displacement of people and submergence of forests and farmland, as in the Narmada project (1); Sedimentation reduces the storage capacity, and dams alter river flow and block fish movement (1); Dams may also cause floods downstream and inter-state disputes, so careful planning and resettlement are needed (1).	1+1+1+1+1
	OR: Development needs water for farming, industry and cities, but overuse harms rivers and aquifers (1); Planning should assess environmental and social costs of projects before they are built (1); Rainwater harvesting, recycling and efficient irrigation reduce demand (1); Affected people must be consulted and compensated (1); Regulation of groundwater and community participation protect resources for the future (1).	1+1+1+1+1

20	Sringaverapura near Allahabad had a sophisticated system to channel the floodwaters of the Ganga (1); During Chandragupta Maurya's time, dams, lakes and irrigation systems were built extensively (1); Evidence of irrigation works has been found in Kalinga (Odisha), Nagarjunakonda (Andhra Pradesh), Bennur (Karnataka) and Kolhapur (Maharashtra) (1); In the eleventh century, Bhopal Lake was one of the largest artificial lakes of its time (1); In the fourteenth century, the tank at Hauz Khas in Delhi was built to supply water to the Siri Fort area (1).	1+1+1+1+1
	OR: Western Himalayas use guls or kuls, channels that carry stream water to terraced fields (1); In the arid and semi-arid Rajasthan, rooftop rainwater is stored in underground tanks (tankas) in Bikaner, Phalodi and Barmer, and khadins and johads hold monsoon run-off on farms (1); In Bengal's flood plains inundation channels were used to carry floodwater to the fields (1); In Meghalaya, a 200-year-old bamboo drip irrigation system carries spring water to plants (1); Each method uses local material and local rainfall and terrain, which is why it suits its region (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 8: **Water Resources (Geography)** | Class 10 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** submerge forests and affect fish and river flow	Environmental cost
2	**(a)** concrete surfaces prevent percolation	Paved surfaces
3	**(d)** rainwater harvesting	Helps percolation
4	**(a)** Both A and R are true and R is the correct explanation of A.	Both true; R explains A
5	**(a)** Both A and R are true and R is the correct explanation of A.	Both true; R explains A
6	**(a)** evaporation	Water goes up as vapour
7	**(d)** rooftop rainwater harvesting	State policy
8	**(d)** displacement of people	Resettlement problems
9	**(d)** traditional rainwater harvesting	Recharge groundwater
10	**(c)** symbolised progress through irrigation and power	They promised development

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	They suit local climate and soil (1); They were developed by communities over generations (1).	1+1
12	They submerge villages and displace people (1); Benefits go to some regions while costs fall on others (1).	1+1
13	Urban demand is high and sources are limited (1); Reuse cuts demand and reduces sewage (1).	1+1
14	Government, industry and citizens all use water (1); Each must save and protect it for future generations (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Water scarcity is a long-term shortage due to overuse and unequal distribution (1); Drought is a temporary lack of rain (1); Scarcity can occur even in high rainfall regions (1).	1+1+1
16	Traditional tanks and kuls give local solutions (1); Modern projects give large supply (1); Together they provide sustainable use (1).	1+1+1
17	Cities grow and demand more water (1); Concrete reduces recharge (1); Pollution adds to scarcity (1).	1+1+1
	OR: Local needs differ (1); Communities can manage resources better (1); It avoids large-scale displacement (1).	1+1+1
18	Glaciers melt and rainfall patterns change (1); Floods and droughts increase (1); Water supply becomes uncertain (1).	1+1+1
	OR: Drip and sprinkler irrigation use less water (1); Crop choice can suit local water (1); Mulching and soil conservation save moisture (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	India has about 4 per cent of the world's renewable water but 16 per cent of its population, and only 2.5 per cent of the world's water is fresh (1); Rising population, intensive irrigation, industrial use and urban demand have increased consumption (1); Over-extraction of groundwater by tube wells lowers water tables, while pollution and unequal access reduce usable water (1); Solutions include rainwater harvesting, efficient irrigation like drip and sprinklers, reuse and recycling of water (1); Community management, regulation of extraction and protection of rivers from pollution complete the answer (1).	1+1+1+1+1
	OR: Irrigation takes the largest share of water, and multiple cropping and Green Revolution crops raised the use of groundwater through tube wells (1); Industries need large quantities of water and discharge polluted effluents, which reduces usable water (1); Urbanisation increases household demand and paves large areas, reducing recharge of groundwater (1); Together they lower water tables and cause wells to dry up (1); Integrated management, recycling and rainwater harvesting are needed to balance the demands (1).	1+1+1+1+1

20	The Narmada Bachao Andolan opposed the Sardar Sarovar Dam because it would displace tribal and other villagers in Gujarat, Madhya Pradesh and Maharashtra (1); Protests also arose against the Tehri dam on the Bhagirathi in Uttarakhand (1); Participants raised questions about resettlement, loss of livelihood and environmental damage (1); They argued that the benefits go to some while the costs fall on local people (1); These movements forced attention to rehabilitation and encouraged the consideration of smaller, local alternatives (1).	1+1+1+1+1
	OR: The government makes laws, builds and manages projects, regulates groundwater and promotes harvesting, as Tamil Nadu did (1); It can enforce pollution control and fair sharing of river water (1); Citizens can save water, repair leaks and avoid wastage (1); Households can harvest rainwater and reuse water (1); Communities can build and maintain structures like johads and kuls, and the combined effort conserves water (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 8: **Water Resources (Geography)** | Class 10 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** tanka - Assam	Tankas are found in Rajasthan
2	**(a)** Both A and R are true and R is the correct explanation of A.	Both true; R explains A
3	**(d)** tanka	A traditional structure of the Thar
4	**(a)** over-extraction of groundwater	Extraction exceeds recharge
5	**(d)** displacement and sedimentation	Costs of dams
6	**(b)** harvesting and reusing water	Conserves for future
7	**(a)** over-extraction and drought	Falling water tables
8	**(b)** Hirakud - Mahanadi	Hirakud is in Odisha
9	**(a)** Both A and R are true and R is the correct explanation of A.	Both true; R explains A
10	**(b)** sharing of river water	Dams and rivers cross states

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Without trees run-off increases (1); Rivers get silted and dry in summer (1).	1+1
12	Rivers cross borders and each state wants more (1); Historical rights and politics complicate agreements (1).	1+1
13	Cities have concrete surfaces that block recharge (1); Harvesting captures run-off and recharges groundwater (1).	1+1
14	They use local materials and simple designs (1); They need no big machinery or compensation costs (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Dams supply power and water to industry (1); They support urban growth (1); Industry then expands near projects (1).	1+1+1
16	Narmada protest was against Sardar Sarovar in Gujarat (1); Tehri protest was against the dam on the Bhagirathi in Uttarakhand (1); Both raised displacement and environmental issues (1).	1+1+1
17	Effluents pollute rivers and groundwater (1); Polluted water is unfit for use (1); Usable supply falls (1).	1+1+1
	OR: Forests hold rain and recharge groundwater (1); Roots reduce run-off (1); Streams flow longer (1).	1+1+1
18	Dams provide irrigation, power and flood control (1); But they displace people and harm ecology (1); So they are important but their costs must be weighed (1).	1+1+1
	OR: Tribunals decide shares between states (1); Their awards are legally binding but sometimes resisted (1); Political will is also needed (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Groundwater irrigates a large part of Indian agriculture through wells and tube wells, especially after the Green Revolution (1); It gave farmers assured water and enabled multiple cropping (1); Extraction beyond the rate of recharge lowers the water table and dries up wells (1); Pumping costs rise, and small farmers cannot afford deeper wells (1); Rainwater harvesting and regulated extraction are needed to recharge it (1).	1+1+1+1+1
	OR: Rainwater harvesting has been practised in India for centuries through tankas, johads, khadins and kuls (1); In modern times, it is revived through rooftop harvesting in Shillong and Gendathur in Karnataka (1); Tamil Nadu made rooftop rainwater harvesting structures compulsory in all houses (1); It recharges groundwater and reduces dependence on distant sources (1); Thus it is both an old tradition and a modern answer to scarcity (1).	1+1+1+1+1

20	Large dams are costly, take years to build, displace people and change river ecology, but supply power and irrigation on a large scale (1); Community systems like johads, tankas and kuls are cheap, quick, use local knowledge and recharge groundwater (1); Large projects cause conflict between states and users, while community systems are managed locally (1); Large dams suffer from sedimentation, whereas small structures are easy to maintain (1); A mixed approach, with small structures for local needs and dams for large demands, is most sustainable (1).	1+1+1+1+1
	OR: Rivers flow through several states, so a dam or extraction in one state affects the downstream state (1); States demand larger shares as demand grows, as in the Krishna-Godavari dispute (1); Disputes between users also arise, such as farmers versus industries (1); They can be resolved through negotiation, tribunals and agreements based on fair sharing (1); Efficient use, conservation and cooperative management reduce conflict (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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